Linear Connections and Geodesics

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Motivation

The Riemanninan generalization of straight lines = **Geodesics**

A curve in Euclidean space is a straight line if and only if its acceleration is zero. To use the same idea on a manifold, we need an intrinsic meaning for the **acceleration of a curve**.

However, for a curve $\gamma : I \to M$, the basic obstruction is that $\dot\gamma(t)$ and $\dot\gamma(t_0)$ lie in different vector spaces,

$$\dot\gamma(t) \in T_{\gamma(t)}M, \qquad \dot\gamma(t_0) \in T_{\gamma(t_0)}M,$$

so they cannot be subtracted without an additional rule for comparing tangent spaces.

Linear Connections

Let $\mathcal{T}(M)$ denotes the space of smooth vector fields on M.

Definition

A **linear connection** on a smooth manifold M is a map

$$\nabla : \mathcal{T}(M) \times \mathcal{T}(M) \longrightarrow \mathcal{T}(M), \qquad (X, Y) \longmapsto \nabla_X Y,$$

satisfying the following properties.

For $f, g \in C^\infty(M)$, $a, b \in \mathbb{R}$, and smooth vector fields,

$$\begin{aligned} \nabla_{fX_1+gX_2} Y &= f\nabla_{X_1} Y + g\nabla_{X_2} Y, \\ \nabla_X(aY_1 + bY_2) &= a\nabla_X Y_1 + b\nabla_X Y_2, \\ \nabla_X(fY) &= (Xf)Y + f\nabla_X Y. \end{aligned}$$

The vector field $\nabla_X Y$ is called the **covariant derivative of Y in the direction X**.

Christoffel Symbols

Let $(E_1, \ldots, E_n)$ be a smooth local frame on an open set $U \subset M$. For every i, j, write

$$\nabla_{E_i} E_j = \Gamma_{ij}^k E_k.$$

The n^3 smooth functions Γ_{ij}^k are called the **Christoffel symbols** of ∇ with respect to the frame.

If

$$X = X^i E_i, \qquad Y = Y^j E_j,$$

then the defining properties of a linear connection imply

$$\nabla_X Y = \left(XY^k + X^i Y^j \Gamma_{ij}^k \right) E_k.$$

Thus the Christoffel symbols completely determine the connection on U.

Christoffel Symbols

Derivation of the local formula

Using the product rule in the second argument and $C^\infty(M)$-linearity in the first argument,

$$\begin{aligned}
\nabla_X Y &= \nabla_X (Y^j E_j) \\
&= (XY^j)E_j + Y^j \nabla_X E_j \\
&= (XY^j)E_j + X^i Y^j \nabla_{E_i} E_j \\
&= (XY^j)E_j + X^i Y^j \Gamma_{ij}^k E_k \\
&= \big(XY^k + X^i Y^j \Gamma_{ij}^k\big) E_k.
\end{aligned}$$

Remark $\nabla_X Y\,|_p$ depends only on the values of Y in a neighborhood of p and the value of X at p.

$$\nabla_X Y\,|_p = \nabla_{X^i \partial_i} Y\,|_p = X^i(p) \nabla_{\partial_i} Y\,|_p$$

Therefore, we can write $\nabla_{X_p} Y := \nabla_X Y\,|_p$

The Euclidean Connection

On $\mathbb{R}^n$ with standard coordinates, define

$$\nabla_X \left(Y^j \partial_j\right) = (XY^j)\partial_j.$$

This is the **Euclidean connection**.

Its Christoffel symbols in the standard coordinate frame all vanish,

$$\Gamma_{ij}^k = 0.$$

Hence the covariant derivative is obtained by differentiating the component functions of Y in the direction X.

Existence and Local Uniqueness

Lemma 4.4

Suppose M is covered by a single coordinate chart $(x^1, \ldots, x^n)$. For any choice of n^3 smooth functions Γ_{ij}^k, the formula

$$\nabla_X Y = \left(X^i \partial_i Y^k + X^i Y^j \Gamma_{ij}^k\right) \partial_k$$

defines a linear connection on M.

Conversely, every linear connection has this form for its Christoffel symbols. Therefore, after the functions Γ_{ij}^k are prescribed in the chart, the corresponding linear connection is unique.

Only the product rule requires a calculation, and it follows from

$$\partial_i(fY^k) = (\partial_i f)Y^k + f\partial_i Y^k.$$

Existence of Linear Connections

Proposition 4.5 Every smooth manifold admits a linear connection.

Proof.
Choose a coordinate cover $\{U_\alpha\}$ of M and a linear connection ∇^α on each U_α. Let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to this cover. Define

$$\nabla_X Y = \sum_\alpha \rho_\alpha \nabla^\alpha_X Y,$$

where each term is extended by zero outside U_α.

The sum is locally finite, and the linearity properties are immediate. For the product rule,

$$\begin{aligned}\nabla_X(fY) &= \sum_\alpha \rho_\alpha \nabla^\alpha_X (fY) \\ &= \sum_\alpha \rho_\alpha \big((Xf)Y + f\nabla^\alpha_X Y\big) \\ &= (Xf)Y + f\nabla_X Y,\end{aligned}$$

because $\sum_\alpha \rho_\alpha = 1$. $\square$

Vector Fields Along Curves

Definitions

Let $\gamma : I \to M$ be a smooth curve. A **vector field along** γ is a smooth map

$$V : I \longrightarrow TM$$

such that $V(t) \in T_{\gamma(t)}M$ for every $t \in I$.

The space of vector fields along γ is denoted by $\mathcal{T}(\gamma)$(e.g.$\dot{\gamma} \in \mathcal{T}(\gamma)$)

A vector field V along γ is called **extendible** if there is a smooth vector field $\widetilde{V}$ on a neighborhood of $\gamma(I)$ such that

$$V(t) = \widetilde{V}_{\gamma(t)}, \quad t \in I.$$

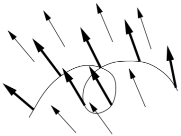


FIGURE 4.4. Extendible vector field.

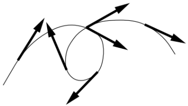


FIGURE 4.5. Nonextendible vector field.

Covariant Derivatives Along Curves

Lemma 4.9

Let ∇ be a linear connection on M and let $\gamma : I \to M$ be a curve. There exists a unique operator

$$D_t : \mathcal{T}(\gamma) \longrightarrow \mathcal{T}(\gamma)$$

satisfying the following properties.

For $a, b \in \mathbb{R}$, $f \in C^\infty(I)$, and $V, W \in \mathcal{T}(\gamma)$,

$$D_t(aV + bW) = aD_tV + bD_tW,$$
$$D_t(fV) = \dot{f}V + fD_tV.$$

If V is extendible and $\widetilde{V}$ is any extension, then

$$D_tV(t) = \nabla_{\dot{\gamma}(t)}\widetilde{V}.$$

The vector field D_tV is called the **covariant derivative of V along γ**.

Covariant Derivatives Along Curves

Uniqueness

The product rule and a bump function on I first show that $D_t V(t_0)$ depends only on the values of V near t_0.

Suppose that $V = W$ on a neighborhood of t_0. Choose a bump function $\varphi \in C^\infty(I)$ such that $\varphi = 1$ near t_0 and $\operatorname{supp}\varphi$ is contained in that neighborhood. Then $\varphi(V - W) = 0$. By the product rule,

$$0 = D_t\big(\varphi(V - W)\big)(t_0) = \varphi'(t_0)\big(V(t_0) - W(t_0)\big) + \varphi(t_0)\big(D_t V(t_0) - D_t W(t_0)\big).$$

Since $V(t_0) = W(t_0)$ and $\varphi(t_0) = 1$, it follows that

$$D_t V(t_0) = D_t W(t_0).$$

Thus $D_t V(t_0)$ depends only on the values of V near t_0.

Covariant Derivatives Along Curves

Uniqueness

Fix $t_0 \in I$ and choose coordinates near $\gamma(t_0)$. Write

$$V(t) = V^j(t)\partial_j$$

for t near t_0. Since each coordinate vector field ∂_j is extendible, the defining properties force

$$\begin{aligned} D_t V(t_0) &= D_t\big(V^j \partial_j\big)(t_0) \\ &= \dot{V}^j(t_0)\partial_j + V^j(t_0)\nabla_{\dot{\gamma}(t_0)}\partial_j. \end{aligned}$$

Thus every admissible operator must have the same value at every t_0. Therefore such an operator is unique if it exists.

Covariant Derivatives Along Curves

Using

$$\dot{\gamma}(t) = \dot{\gamma}^i(t)\partial_i$$

and

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k,$$

we obtain the coordinate formula

$$D_t V(t) = \left(\dot{V}^k(t) + V^j(t)\dot{\gamma}^i(t)\Gamma^k_{ij}(\gamma(t))\right)\partial_k.$$

The first term differentiates the components of V. The Christoffel-symbol term corrects this ordinary derivative for the variation of the coordinate frame along the curve.

Covariant Derivatives Along Curves

Existence

If the image of γ lies in a single coordinate chart, define $D_t V$ by

$$D_t V = \left(\dot{V}^k + V^j \dot{\gamma}^i \Gamma_{ij}^k(\gamma) \right) \partial_k.$$

A direct calculation verifies linearity, the product rule, and agreement with ∇ for extendible fields.

For a general curve, cover its image by coordinate charts and use this formula on the corresponding subintervals of I. On overlaps, the locally defined operators satisfy the same three characterizing properties. By uniqueness they agree, so they combine to give a globally defined operator D_t along γ. $\square$

Acceleration and Geodesics

Let M be equipped with a linear connection ∇, and let $\gamma : I \to M$ be a smooth curve.

The **acceleration** of γ is the vector field along γ defined by

$$D_t\dot{\gamma}.$$

The curve γ is called a **geodesic with respect to** ∇ if

$$D_t\dot{\gamma} \equiv 0.$$

This definition depends on the chosen linear connection.

Acceleration and Geodesics

For the Euclidean connection on $\mathbb{R}^n$,

$$D_t\dot{\gamma}(t) = \ddot{\gamma}^k(t)\partial_k.$$

Therefore the geodesic equation is

$$\ddot{\gamma}^k(t) = 0, \qquad k = 1, \ldots, n.$$

Its solutions are

$$\gamma(t) = a + bt, \quad a, b \in \mathbb{R}^n$$

Hence the Euclidean geodesics are exactly the straight lines with constant-speed.

The Geodesic Equation

Let $\gamma(t)$ have coordinate functions

$$\gamma(t) = \big(x^1(t), \dots, x^n(t)\big).$$

Since

$$\dot{\gamma}(t) = \dot{x}^j(t)\partial_j,$$

the coordinate formula for covariant differentiation gives

$$D_t\dot{\gamma} = \Big(\ddot{x}^k + \dot{x}^i\dot{x}^j\Gamma^k_{ij}(\gamma(t))\Big)\,\partial_k.$$

Thus γ is a geodesic if and only if

$$\ddot{x}^k(t) + \dot{x}^i(t)\dot{x}^j(t)\Gamma^k_{ij}(\gamma(t)) = 0, \qquad k = 1, \dots, n.$$

Existence and Uniqueness of Geodesics

Theorem 4.10

Let M be a manifold with a linear connection. For every $p \in M$, $V \in T_pM$, and $t_0 \in \mathbb{R}$, there exist an open interval I containing t_0 and a geodesic

$$\gamma : I \longrightarrow M$$

satisfying

$$\gamma(t_0) = p, \qquad \dot{\gamma}(t_0) = V.$$

Any two geodesics with the same initial position and initial velocity agree on their common domain.

Existence and Uniqueness of Geodesics

Proof.
Choose local coordinates near p and set

$$v^k(t) = \dot{x}^k(t).$$

The second-order geodesic equation is equivalent to the first-order system on $U \times \mathbb{R}^n$,

$$\begin{aligned} \dot{x}^k(t) &= v^k(t), \\ \dot{v}^k(t) &= -v^i(t)v^j(t)\Gamma^k_{ij}(x(t)). \end{aligned}$$

The right-hand side is smooth in (x, v). The standard existence and uniqueness theorem for first-order ordinary differential equations therefore gives a unique local solution with initial data

$$x(t_0) = p, \qquad v(t_0) = V.$$

Its x-component is the required local geodesic.

Existence and Uniqueness of Geodesics

Proof (continued).
Suppose γ and σ are geodesics with the same initial data. Local uniqueness for the first-order system implies that they agree near t_0.

If they agree up to a time β in their common domain, then continuity gives

$$\gamma(\beta) = \sigma(\beta), \qquad \dot{\gamma}(\beta) = \dot{\sigma}(\beta).$$

Applying local uniqueness again at β extends the interval on all of I. □

Taking the union of all intervals carrying the same initial-value solution yields a **unique maximal geodesic** for $p \in \mathcal{M}$ and $V \in T_p\mathcal{M}$, denoted by γ_V.

Parallel Vector Fields

A vector field V along a curve γ is called **parallel along** γ if

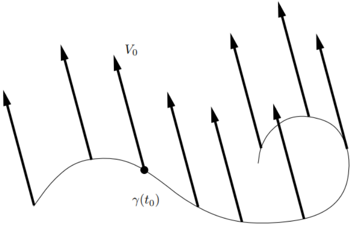
$$D_t V \equiv 0.$$

(A curve is a geodesic precisely when its velocity field is parallel).

For the Euclidean connection, the parallel equation is

$$\dot{V}^k(t) = 0,$$

so parallel vector fields are exactly those with constant standard-coordinate components.



Parallel Translation

Theorem 4.11

Let $\gamma : I \to M$ be a curve, let $t_0 \in I$, and let $V_0 \in T_{\gamma(t_0)}M$. There exists a unique parallel vector field V along γ satisfying

$$V(t_0) = V_0.$$

This field is called the **parallel translate of V_0 along** γ.

The result depends on both the chosen connection ∇ and the chosen curve γ.

Parallel Translation

Proof.
If $\gamma(I)$ lies in one coordinate chart and

$$V(t) = V^k(t)\partial_k,$$

then the equation $D_t V = 0$ is

$$\dot{V}^k(t) = -\dot{\gamma}^i(t)\Gamma_{ij}^k(\gamma(t))V^j(t).$$

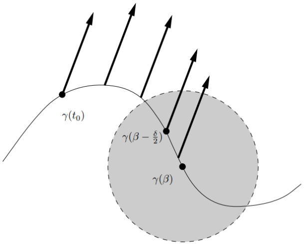
This is a linear system of ordinary differential equations for the component functions $V^1, \ldots, V^n$.

A smooth linear initial-value problem has a unique solution on the entire interval I. Unlike a general nonlinear ODE, a linear solution cannot blow up at a finite interior time of its coefficient interval. Therefore the parallel translate exists uniquely on all of I when one chart contains the curve.

Parallel Translation

Proof for a general curve.

Let β be the supremum of times to which the parallel field with the prescribed initial value has been uniquely extended. If β were an interior point of I, choose a coordinate chart containing a short segment $\gamma((\beta - \delta, \beta + \delta))$.



Solve the linear parallel equation in this chart using the already constructed value at $\beta - \delta/2$ as initial data. By uniqueness, the new solution agrees with the old one on their overlap and extends it past β, a contradiction.

The same argument to the left of t_0 gives a unique parallel field on all of I. □

The Parallel Translation Operator

For $s, t \in I$, define

$$P_{st} : T_{\gamma(s)}M \longrightarrow T_{\gamma(t)}M$$

by

$$P_{st}(V_0) = V(t),$$

where V is the unique parallel field satisfying $V(s) = V_0$.

Linearity of the parallel ODE and uniqueness imply

$$\begin{aligned} P_{ss} &= \mathrm{Id}, \\ P_{tu} \circ P_{st} &= P_{su}, \\ P_{st}^{-1} &= P_{ts}. \end{aligned}$$

Thus P_{st} is a **linear isomorphism between tangent spaces** at different points of the curve.

Recovering Covariant Differentiation

Parallel translation allows us to compare $V(t)$ with $V(t_0)$ inside one vector space. The covariant derivative along γ is recovered by

$$D_t V(t_0) = \lim_{t \to t_0} \frac{P_{t_0 t}^{-1} V(t) - V(t_0)}{t - t_0}.$$

This formula makes precise the statement that a connection “connects” nearby tangent spaces.

Parallel Frames

Fix $t_0 \in I$ and choose a basis

$$(e_1, \dots, e_n)$$

of $T_{\gamma(t_0)}M$. For each i, let E_i be the parallel translate of e_i along γ. Then

$$E_i(t) = P_{t_0 t}(e_i).$$

Because $P_{t_0 t}$ is an isomorphism,

$$(E_1(t), \dots, E_n(t))$$

is a basis of $T_{\gamma(t)}M$ for every $t \in I$.

The collection $(E_1, \dots, E_n)$ is called a **parallel frame along** γ. Every curve admits a parallel frame once a linear connection and an initial basis are chosen.

Recovering Covariant Differentiation

Proof.

Choose a parallel frame $(E_1, \ldots, E_n)$ along γ and write

$$V(t) = V^i(t)E_i(t).$$

Since $D_tE_i = 0$,

$$D_tV(t_0) = \dot{V}^i(t_0)E_i(t_0).$$

Also, parallel translation sends $E_i(t_0)$ to $E_i(t)$, so

$$P_{t_0 t}^{-1}V(t) = V^i(t)E_i(t_0).$$

Therefore

$$\begin{aligned}
\lim_{t\to t_0} \frac{P_{t_0 t}^{-1}V(t) - V(t_0)}{t - t_0} &= \lim_{t\to t_0} \frac{V^i(t) - V^i(t_0)}{t - t_0} E_i(t_0) \\
&= \dot{V}^i(t_0)E_i(t_0) \\
&= D_tV(t_0).
\end{aligned}$$

□